

Slesa Adhikari

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WORK EXPERIENCE

NASA / University of Alabama in Huntsville (UAH)

Jun. 2020 – Present

Computer Scientist / Software Engineer (Promoted through Level VI)

Huntsville, AL

- Lead engineer/contributor for multiple cloud-native Earth science platforms supporting cataloging, visualization, and analysis of large geospatial datasets (e.g., U.S. Greenhouse Gas Center, VEDA, MAAP).
- Designed and implemented high-performance geospatial visualization UIs using React and modern WebGL-based tooling (Mapbox GL, Deck.gl, Cesium), enabling interactive exploration of large-scale Earth science datasets.
- Designed and operated AWS-based geospatial data pipelines using STAC metadata, Cloud Optimized GeoTIFFs (COGs), tiling services, and serverless/containerized workloads.
- Implemented infrastructure-as-code (AWS CDK, Terraform) and CI/CD pipelines (GitHub Actions) to automate deployments across multiple environments.
- Developed and maintained APIs and data services (FastAPI, Django; REST/GraphQL) backed by Postgres, DynamoDB, and Neo4j.
- Led technical design reviews, mentored engineers, and partnered with NASA stakeholders on architecture and roadmap decisions.
- Supported national-scale platforms serving thousands of users and large-scale datasets.
- Reduced deployment friction and environment drift through standardized IaC and CI/CD workflows.
- Contributed to award-winning projects recognized by NASA Marshall Innovation Team Awards.

Saano Labs LLC

Nov. 2025 – Present

Founder, CEO | Nov 2025 – Present

Huntsville, AL

- Develop and operate Earth Science-based data systems for cataloging, ingestion, and analytics on large-scale geospatial datasets for the IEEE GRSS community.

University of Alabama in Huntsville (UAH)

Aug. 2018 – May 2020

Graduate Research Assistant / Software Engineer | May 2019 – May 2020

Huntsville, AL

- Processed large Earth Science datasets using NumPy, xarray, and zarr.
- Optimized 3D web visualization performance for Cesium-based applications.
- Built and deployed serverless REST APIs on AWS Lambda using Terraform.

Graduate Teaching Assistant | Aug. 2018 – May 2019

- Supported instruction, labs, grading, and tutoring for undergraduate computer engineering courses.

Logic Information System Nepal

Jan. 2016 – May 2017

Associate Software Developer | Nov. 2016, May 2017

Kathmandu, Nepal

- Built data warehousing, reporting, and archiving solutions.
- Developed ETL and automation scripts supporting enterprise reporting pipelines.

Software Engineering Intern | Jan. 2016, Sep. 2016

- Developed a VR prototype to reconstruct partial 3D scenes from 2D images.

EDUCATION

University of Alabama in Huntsville

May, 2020

Master's in Computer Science

Huntsville, AL

- **GPA: 3.875**
- *Thesis:* Detecting Periodic Action Patterns in Videos | AAUW International Master's Degree Fellowships

- **Percentage: 78.6%**
- Full merit-based scholarship; Top 10 of ~700 students | Ncell Excellence & Scholarship Awards

SKILLS, INTERESTS & AWARDS

- **Technologies:** Python, FastAPI, JavaScript, React, Deck.gl, Mapbox GL, WebGL, AWS (S3, Lambda, ECS, Step Functions, API Gateway, CloudFront, RDS, DynamoDB, Cognito, MWAA), CDK, Terraform, Docker, GitHub Actions
 - *Geospatial:* STAC, Tiler, COG, rasterio, xarray, zarr, shapefiles, mapping services (WMTS/WMS/XYZ), ArcGIS, QGIS, GDAL
- **Skills:** System design; technical decision-making; cross-team collaboration; mentoring engineers; roadmap planning
- **Interests:** Making art (comics, painting, sketching); interior design; 3D modeling & 3D printing; smart home automation; DIY building & fabrication; traveling; U.S. national parks; sustainability & eco-conscious living; plants, gardening, and native plant ecology.
- **Awards:**
 - NASA Comet Award (2023)
 - NASA Marshall Innovation Team Awards (2021, 2021)
 - AAUW International Fellowships (2018, 2019)